

Supporting Transdisciplinary Research: Combining Design Research with Interactive Research

Kristina SÄFSTEN ^{a,1}, Fredrik ELGH ^a, Roland STOLT ^a, Gary LINNEUSSON ^b, Joel JOHANSSON ^{a, c}

^a Department of Product development, Production and Design, School of Engineering, Jönköping University, Sweden

^b Department of Intelligent Production Systems, School of Engineering Science, University of Skövde, Sweden

^cProduct Development, Thule Sweden AB, Sweden

Abstract. In this paper, a work procedure developed to support the collaboration and progress in the research project *IDEAL – Integrated product and production platforms supporting agile and demand-driven industrial product realisation* – is presented and its functionality discussed. The research project involved in total 13 researchers, five manufacturing companies and one software supplier. The research project was organised in four sub-projects, covering various aspects of a joint research question. The project started in April 2020 and ended in January 2024. The work procedure, called the ‘IDEAL work procedure’, was developed based on the overall principles from interactive research combined with the framework for design research methodology (DRM). The developed work procedure provided a structure for the project, connecting the four subprojects, and thereby supported fulfilment of the joint research question. During the research project, the functionality of the ‘IDEAL work procedure’ has been assessed in different ways, both in terms of how it was perceived and to what extent the planned results have been achieved. During the project we have carried out workshops to follow-up on the progress and the work procedure. In addition, follow-up interviews have been conducted with participants from involved companies. The results from the different assessment activities are synthesised and presented in this paper. To expand the applicability of the ‘IDEAL work procedure’, the potential of the procedure to support transdisciplinary research is elaborated on in this paper.

Keywords. transdisciplinary engineering, transdisciplinary research, collaborative research, learning process, industry-academia collaboration

Introduction

Current societal challenges increasingly require radical new solutions developed in collaboration between academia and industry. However, due to distinct logics, incentives, and timescales, a divide exists between academia and industry [1]. Efforts are needed to combine practical relevance and academic contributions, a challenge associated with applied fields of research such as engineering and management [2, 3]. To illustrate the relation between industry and academia, “the gap” as a metaphor has commonly been

¹ Corresponding Author, Email: kristna.safsten@ju.se.

used [4]. Today, with the progress made in terms of, for example, various collaborative research approaches, “the interface” is suggested as a more appropriate metaphor as a means, or location, of interaction between two systems (industry and academia) [5]. In Sweden, one of the governmental funding bodies, the Knowledge Foundation, emphasises co-production, which means that projects must be designed and implemented in close collaboration between academia and the business sector [6]. In European research at large, collaboration between academic and non-academics have been widely promoted [7]. Knowledge across different fields, technologies, contexts, and scientific disciplines need to be combined, across theory and practice, which calls for strengthened co-productive research approaches [8].

There is no doubt that research benefit from being carried out in close collaboration between different parties, involving both industry and academia, and transdisciplinary research has been pointed out as supportive when solving complex problems [9]. There are, however, several critical aspects related to the co-creation of knowledge between industry and academia that need to be considered, including the process of building partnerships between academic researchers and practitioners, the nature of the collaborative research process, and the challenges involved in this way of working from the point of view of the participants involved [10].

In this paper, a work procedure developed to support the collaboration and progress in the research project IDEAL is presented and its functionality discussed. To expand the applicability of the ‘IDEAL work procedure’, the potential of the procedure to support transdisciplinary research is elaborated on this paper. We specifically investigate to what degree problem framing, integration and implementation of results could be supported, since these are challenges pointed out for transdisciplinary research [7]. We argue that a combination of knowledge, related to collaborative research and industry-academia collaboration, may be a fruitful avenue to strengthen transdisciplinary research.

1. Collaborative and transdisciplinary research

In this section, different perspectives on collaborative research is presented. Literature on different disciplinary concepts focus on who is to be integrated and what is to be achieved by means of integration, rather than how integration is to be carried out and how the expected results are achieved. By combining previous knowledge of different disciplinary concepts with knowledge of different collaborative research approaches a foundation for the reflection of the ‘IDEAL work procedure’ is created.

There are many terms to describe interaction between different disciplines, for example interdisciplinarity, multidisciplinary and transdisciplinarity. The lack of common terminology does however make the landscape a bit vague, which causes confusion in communication and hinders consistent development of practices [9, 11]. As a starting point, disciplinary research can be defined as research carried out within one academic discipline with the goal to create knowledge relevant for that specific discipline [11]. The borders of a discipline are however not always clearcut and easy to define, disciplines are dynamic and varies with field of research, institutions and types of knowledge. Some borders are harder to cross than others, sometimes borders between sub-disciplines might even be the most challenging [11].

To differentiate different disciplinary concepts, a distinction related to the participants and the level of integration has been suggested, see Figure 1 [11]. Multidisciplinary research involves multiple academic disciplines, where each discipline

formulate goals, under a thematic umbrella [11]. Some cooperation take place across the involved disciplines, but the knowledge developed is discipline specific. When the level of integration is low, work in involved disciplines are carried out in parallel. Interdisciplinarity also involves multiple disciplines.

Figure 1. Degree of integration and participants in different disciplinarity concepts [11].

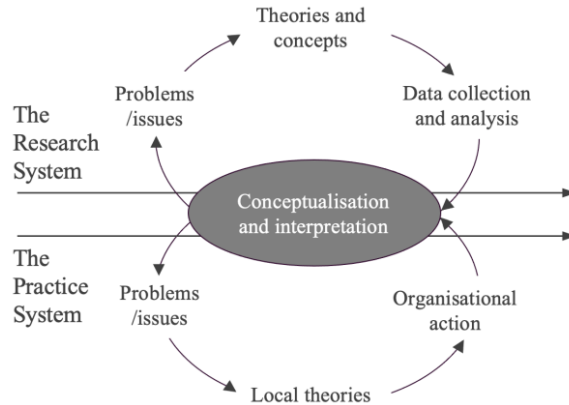
	<i>Low level of integration</i>	<i>High level of integration</i>
<i>Academic and non-academic participants</i>	Participatory (may not be research)	Transdisciplinary research
<i>Academic participants</i>	Multidisciplinary research	Interdisciplinary research

The main differences compared to multidisciplinary research is that a joint goal is formulated, and the knowledge developed is integrated, not discipline specific. Lastly, transdisciplinary research implies that both disciplinary borders and academic/non-academic borders are crossed. A joint goal is formulated, and the research is carried out in collaboration between different academic disciplines and non-academic participants [11]. When collaboration across academic and non-academic participants are discussed, a new mode of knowledge production is required, i.e., creating knowledge together with the stakeholders [7]. Similarly as with interdisciplinary research, in transdisciplinary research integrated knowledge is developed, combining knowledge from different disciplines. In addition, transdisciplinary research is described to be problem focused, addressing real world problems [9]. New knowledge is developed in both academia and among the involved non-academic participants (the society) [11].

Collaboration between industry and academia is commonly advocated as core in applied research, such as engineering and management. Over the recent decades, there has been a noticeable transformation in the dynamics between the academic and industrial sectors, moving from a sponsorship model to a more collaborative partnership approach. The emphasis has shifted from conducting research solely for the purpose of creating practical knowledge handed over to practitioners, to a joint and cooperative process where usable knowledge is co-created through collaboration [10]. A plethora of research approaches are suggested, describing or prescribing how this collaboration could or should be performed to reach the required scientific quality targets and at the same time support development in, for example, the manufacturing industry. Some examples of collaborative research approaches are collaborative management research, action research and interactive research [12-14]. The interactive research approach emphasises that two organizational systems are involved, the research system and practice system, see Figure 2, thus explicitly addressing both the industrial and academic perspectives [3, 15].

The activities in the research system are assumed to be informed by scientific theories and concepts, whereas activities in the practice system, i.e., organizational problem-solving actions, are assumed to be informed local theories based on previous research and/or practical experience [13]. At the core of the interactive research approach is the joint conceptualisation and interpretation of the research object and the co-creation of new knowledge, which are then integrated as cognitive input into subsequent cycles of problem-solving activities and research.

Figure 2. The interactive research approach, with the interacting research system and practice system [15, p. 352].



2. Method and material

In this section the development of the 'IDEAL work procedure' is described together with the activities to evaluate its functionality. Before that, the set-up of the IDEAL-project is briefly described to contextualize the work procedure and its application.

2.1. The research project IDEAL

Between April 2020 and January 2024 the research project 'Integrated product and production platforms supporting agile and demand-driven industrial product realisation (IDEAL)' has been an essential part of the work task for several researchers at School of Engineering, Jönköping University, Sweden. The IDEAL-project was funded by the Knowledge Foundation, a governmental funding agency in Sweden, advocating co-production, which means close collaboration between academia and the business sector [6]. The research project was a so called 'synergi project', which required that a joint core question was formulated that motivate the research. In the IDEAL-project this question was: How can integrated product and production platforms support agile and demand driven product realisation? The core of the 'IDEAL-project' evolved during a period of six months where the academic partners secured the scientific relevance and the companies expressed industrial need, expectations and involvement in Letters of Intent. Workshops were organised followed by additional meetings and contacts, to facilitate the development of the joint core question of the synergy project.

To cover various aspects of the core question the research project was organised in four sub-projects. Two of the sub-projects were targeting platform planning and development (product and production respectively) and two sub-projects addressed means for integration (technical and organisational respectively), addressing the joint research question from different perspectives [16].

The research project involved non-academic partners, more specifically five industrial partners and one software supplier. Among the industrial partners, three were from the traditional mechanical engineering industry and two from the construction industry. The rationale for the composition of the group of industrial partners was the assumption that there was a learning potential between the different industry sectors. Traditional mechanical engineering industry was assumed to be more advanced in terms

of platform planning and development, whereas the selected house builders had capability of delivering customer specific products in low volumes. The software supplier was selected based on their specialisation in software integrating product and production development. The non-academic partners are denoted Company Armature, Outdoor, Transportation, Housing, Operations and InfoHub, see Table 1.

Table 1. Non-academic partners engaged in the IDEAL-project.

Company	Type of product	Type of business, location, etc.	Number of employees at local unit (2023)
Armature	Indoor and outdoor lighting systems	Part of larger group, local product unit included.	452
Outdoor	Outdoor power products for professionals and consumers	Part of larger global group, local unit included	2 417
Transportation	Products for active lifestyle and automotive	Group with global production, local unit included	2600
Housing	Industrialised house building for the private market	Part of larger group, local unit included	900
Operations	Industrialised house building for the private market	Part of larger group, local unit included.	Closed in 2023. In 2021, 110 employees.
InfoHub	Provider of PLM solutions	Small privately owned company	8

During the project design and implementation, one academic institution and in total 14 academic participants were involved. An overall project manager was appointed. Sub-project managers were appointed for each sub-project, of which one was the overall project manager. In each sub-project, additional researchers were engaged. At the start of the project, four new doctoral students were recruited, one related to each sub-project (of which three completed a degree). The expertise in the project team was on product platforms, production systems, product development, production development, simulation, digitization, automation, systems theory, and knowledge integration.

To coordinate the project content and implementation, a steering group with representatives from the industrial partners was formed. Meetings were held at least four times a year, led by the project manager (the sub-project managers attended). In addition, a project management team was formed, including the sub-project managers, with weekly or biweekly pulse meetings. Each sub-project had their own set-up. To create joint understanding and shared visions among all researchers, regular seminars were held in the research group. During the first 2,5 year of the project, different topics related to the project were brought up at the seminars and scientific papers were jointly read and discussed. During the last year of the IDEAL-project, the seminars mainly focused on results from each sub-project and fulfilment of the overall synergy objectives. In addition, synergi workshops including representatives from all participating companies were held at three occasions. The purpose of the initial workshop was to start-up the project. This was at the verge of the pandemic (spring 2020) and therefore this workshop was held on Microsoft Teams. The second workshop was held in May 2022, combined with a study visit at the hosting industrial partner (Company Transportation). The final joint workshop was held at the university, in December 2023.

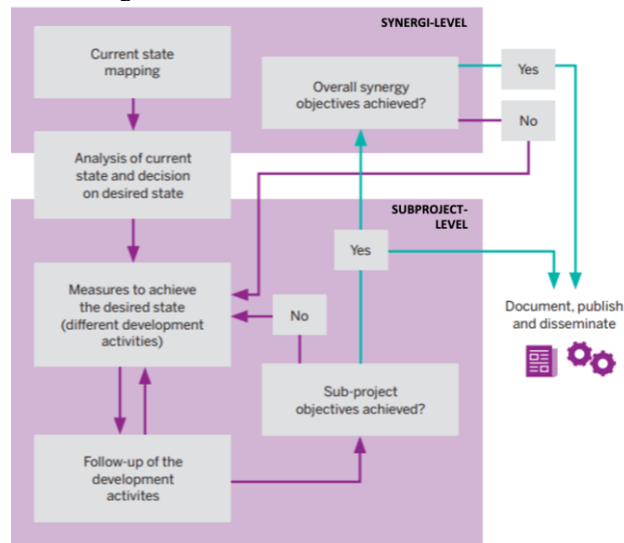
2.2. The 'IDEAL work procedure'

The 'IDEAL work procedure' was developed as part of the design of the IDEAL project. The foundation was the overall principles from interactive research [13, 15] combined with the structure in the framework for design research methodology (DRM) [17]. The 'IDEAL work procedure' has five main stages (corresponding DRM stages are mentioned in brackets):

1. Current state mapping (Research clarification, RC)
2. Analysis of current state and decision on desired state (Descriptive study I, DSI)
3. Implement different development activities (Prescriptive study, PS)
4. Follow-up of the development activities (Descriptive study II, DSII)
5. Follow-up of sub-project and synergi project fulfilment (Descriptive study II, DSII)

The logic between the different stages, and their position at synergi-level and at subproject-level is visualized in Figure 3.

Figure 3. Visualisation of the 'IDEAL work procedure'.



The implementation of the 'IDEAL work procedure', followed the five main stages described, visualised in Figure 3. Initially, focus was on the current state mapping (RC) and the subsequent analysis and decision on the desired state (DS I). The current state mapping was carried out through interviews, combined with document studies. A structured interview guide was created, capturing relevant questions from each sub-project. Each interview was carried out of two researchers, most often representing two of the four subprojects, via Microsoft Teams. In total 53 interviews, with an average time of 1 h 30 min per interview, were carried out in the six companies. Data analysis was carried out in smaller groups of researchers, where all subprojects were represented. As a result, the current state for each company was compiled and thereafter validated with the company representative. To develop a desired state for each company and identify common development activities, workshops were arranged, one for each company and one with all companies, where the results from the current state was presented and discussed, and potential development activities identified. All companies also had internal meetings before and between the workshops. In total more than 70 development

activities were suggested, condensed by the researchers into 13 larger initiatives, agreed upon by the industrial partners. Thereafter the implementation of the different development activities (PS) commenced, carried out on sub-project level, see Figure 3. Follow-up of the development activities (DSII) was continuously carried out by each sub-project.

2.2.1. Activities to assess the functionality of the 'IDEAL work procedure'

To anchor the 'IDEAL work procedure' among the academic researchers in the project team, a workshop was arranged where the visual appearance was assessed with support from a framework including cognitive, communicative and collaborative aspects of visualisation [18]. During the workshop, nine of the total 13 researchers participated (all doctoral students was not yet recruited). The IDEAL-project was assessed during the second workshop in May 2022. During the workshop, the possibility of reaching the goals in the ongoing development activities, sub-projects, and in the IDEAL-project was discussed among industrial and academic partners. During the last semester of the research project, the progress at each company was revisited and the goal-fulfilment assessed. During the concluding result conference, in December 2023, a workshop with focus on the co-production in the IDEAL-project was arranged. During this workshop, the academic and the industrial partners discussed the co-production during the different phases of the IDEAL-project. In addition, the hosting academic institution assessed the co-production. The assessment was conducted through digital surveys to participating industrial partners and the project manager at three occasions during the project.

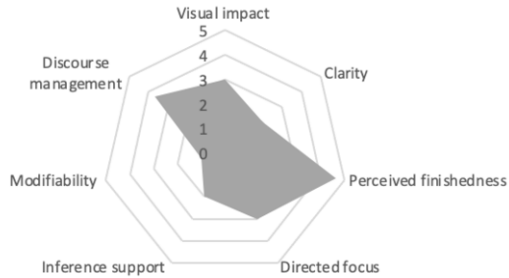
3. Results

In this section the perceived functionality of the 'IDEAL work procedure' is presented from different perspectives. As a starting point, the results from the initial anchoring activity are presented. Thereafter the 'IDEAL work procedures' ability to frame the problem, its ability to support integration, and finally its ability to support implementation of results are described.

3.1. Visualisation of the 'IDEAL work procedures'

The visualisation of the 'IDEAL work procedure' was assessed among the academic researchers. The assessment included communicative, cognitive and collaborative dimensions of visualisation, including: visual impact, clarity, perceived finishedness, directed focus and inference support, and finally modifiability and discourse management [18], see Figure 4. Each aspect was assigned a value between 1 (low) and 5 (high). The extent to which the visualisation of the 'IDEAL work procedure' was perceived as attractive and facilitating attention and recall, was average (visual impact = 3). The work procedure was not perceived as self-explanatory (clarity = 2), but it was perceived as finished (perceived finishedness = 5). The visualisation drew attention to several items, i.e., the directed focus was average (3) but the visualisation as such did not generate new insights (inference support = 2). The extent to which the work procedure provided control over the discussion and workflow was a fairly good (discourse management = 4) but possibility to alter in response to the dynamics of a situation was as expected limited (modifiability = 1).

Figure 4. Assessment of the visualisation of the 'IDEAL work procedure'.



The results indicated that the 'IDEAL work procedure' was perceived as complete and to have a fairly good potential to keep control over the discussion and workflow. The low grade on clarity (2) highlighted the need for clear communication of the intention with the work procedure to allow for shared understanding in the project team, which thereby was important input during project start up.

3.2. Framing the problem and supporting integration

When the project was designed, the overall research question (How can integrated product and production platforms support agile and demand driven product realisation?) was general enough to be of interest to all involved companies. The subsequent current state mapping (RC) and analysis of current state and decision on desired state (DSI) provided a foundation for the upcoming development activities. During the concluding result conference, the industrial participants emphasised the value of the initial interviews and workshops, and the joint creation of development activities. Although extensive and relatively time-consuming, this stage was perceived to contribute to improved relations between the researchers and the industrial participants. The result was a limited number of development activities (13), formulated jointly by the researchers and the industrial participants. The procedure settled the commitment from industry and served as a second agreement on the project intentions. During the project, we regularly reported on the progress related to the development activities, and thereby all the involved participants were continuously reminded about the overall project purpose and the agreed activities. Workshops, study visits, and so-called knowledge sharing sessions were described as very beneficial for integration between the involved participants.

3.3. Support for implementation of results

The implementation of the development activities was carried out within the sub-projects. Teams with researchers and specialists from the companies were organised for the activities. The actual execution of the activities was adapted to the specific development activity at hand and included interviews, workshops, knowledge sharing sessions, document analysis, method and demonstrator development, prototype testing and evaluation. The project results included new insights and knowledge as well as methods and demonstrators related to platform development and organisational and technological integration. For example, a method that support requirements management in the interface between product development and production has been developed, a method to analyse the long-term capabilities of a production system, a bill of material master and a method for boundary object assessment. The results from hosting organisations regular

assessment of perceived co-production capability shows that the industrial partners perceived several different results from the project, both in terms of specific methods and tools to be used, but also in terms of inspiration and learnings valuable for their continued development work. In terms of academic results, two journal articles and 23 conference papers have been published, and work is ongoing with additional eight articles. In addition, three licentiate theses have successfully been defended.

4. Discussion and conclusion

The 'IDEAL work procedure' provided a framework for the IDEAL-project, connecting the four subprojects, and supported fulfilment of the joint core question. To expand the applicability of the 'IDEAL work procedure', the potential of the procedure to support transdisciplinary research was elaborated on in this paper. Transdisciplinary researchers must understand the complexity of a problem, consider diverse perspectives, connect abstract and case-specific knowledge, and foster knowledge and practices that advance the common good in order to be effective in problem-solving endeavors [7]. With this as background, it has been pointed out that transdisciplinary researchers need to frame the problem, to integrate various perspectives, and implement results in practice [7]. The question raised in this paper was to what degree the 'IDEAL-work procedure' can support the requirements put on the transdisciplinary researcher.

The initial stages of the 'IDEAL work procedure', current state mapping (RC) and analysis of current state and decision on desired state (DSI) contributed to both framing of the problem and to the integration between the involved participants. Working across disciplinary and cultural borders requires communication of information and knowledge across borders. Capability of different disciplines to interpret the same object, expressed in their domain languages is required to succeed with transdisciplinary knowledge production [19]. The concept boundary object refers to objects that enable individuals with diverse specialisations to bridge across their specialisations and integrate knowledge [20, 21]. The 'IDEAL work procedure' connecting the different academic disciplines and the academia-industry perspectives. Based on the results from the initial assessment of the visualisation, it was possible to clarify the purpose and the intended way of implementing the 'IDEAL work procedure'. Another important element that have contributed to the transdisciplinarity in the IDEAL-project, was the 13 development activities formulated as a result of the initial stages of the 'IDEAL work procedure'.

The 'IDEAL work procedure' also supported result implementation in practice, the third requirement on a transdisciplinary researcher. The jointly executed development activities allowed the industrial partners to be part of the development of methods and tools, and thereby monitor its relevance. Workshops, study visits, and other activities within the research project was perceived to inspire and provide learnings valuable for their continued development work. Traditionally, the goal of transdisciplinary research extends beyond merely solving tangible problems; it also aims to enrich the collective knowledge within the community of practice, encompassing the research community and the industrial community [19]. To sum up, the 'IDEAL work procedure' is an example of how transdisciplinary research can be supported by methods and tools developed to support co-production and industry-academia collaboration, without explicitly addressing transdisciplinary research. To develop the area of transdisciplinarity research, collaboration with researchers focusing on how to conduct collaborative research seems to be a fruitful way forward.

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